



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University

NAAC Accredited with 'A+' Grade & An ISO 9001: 2015 Certified Institution

NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Computer Science and Engineering

Academic Year: 2025 - 2026 (Odd Semester)

Name of the Faculty member : Mrs.S.Shunmuga Priya, AP/CSE
Degree, Semester & Branch, Sec : B.E, V & CSE – 'B'
Course Code & Title : CCS375 WEB TECHNOLOGIES
Date & Time : 24.10.2025 & 12.35 PM to 1.00 PM

Innovative Practice V - Description

- Unit / Topic: Unit IV / PHP - database connectivity
- Course Outcome: CO4
- Topic Learning Outcome : TLO4a
- **Activity Chosen: Project Based Learning**

Justification

- This innovative practice was assigned to strengthen students' conceptual and practical understanding of database connectivity in PHP through a Project-Based Learning (PBL) approach. In the Web Technology curriculum, there is no separate laboratory component dedicated to PHP programming. Traditionally, PHP and database operations are taught through brief code snippets or lecture explanations, making it difficult for students to visualize how web applications interact with databases in real-time.
- To bridge this gap, students were engaged in a real-world mini project "Student Profile Management System" where the students collaboratively designed a PHP-based web application that connects to a MySQL database. This approach encouraged students to explore the complete data flow between client, server, and database while performing real CRUD operations.

Time Allotted for the Activity: 25 Minutes

Implementation:

- In this Project-Based Learning activity, one team consisting of three students namely Surya prabha J, Nandhini S, Panimalar N was formed based on their willingness and interest in learning PHP and backend connectivity concepts. The project title, "Student Profile Management System," was assigned two weeks before the demonstration, allowing adequate time for planning, coding, and testing.
- During the first week, the team analyzed the requirements and designed the database schema using MySQL, including tables for student details, login credentials, and academic information.

- In the second week, the students developed the front-end interface using HTML and CSS and implemented PHP scripts to establish database connectivity, perform CRUD (Create, Read, Update, Delete) operations, and handle form submissions. The system was tested for data retrieval accuracy and user validation. Screenshots of the staff dashboard and student profile details are shown in Figure 1.
- The project was finally demonstrated by the team using a projector in the classroom, that shows the working model of the “student profile management system” and effectively demonstrating the concept of database connectivity in PHP through a practical approach as shown in figure 2.

CO – PO / PSO mapping:

CO →	PO1	PO2	PO3	PO9	PO10	PO11	PSO1
CO4	3	3	3	1	2	2	3

PO-PSO Mapped:

Innovative Practice	PO1	PO2	PO3	PO9	PO10	PO11	PSO1
	3	3	3	1	2	2	3
Justification for Correlation	Students apply fundamental programming knowledge in PHP, MySQL, HTML, and CSS to design and implement a functional web-based system.	Students analyze system requirements, design the database schema, and develop the backend logic to solve real-world data management problems.	Students design, develop, and test a complete web application integrating frontend and backend components effectively.	Working in a team of three, students collaborate effectively, share tasks, and coordinate to complete the project within the deadline.	Students present and demonstrate their working model confidently, explaining technical implementation and design decisions.	Students demonstrate project management and self-learning abilities by planning timelines, dividing responsibilities, and ensuring successful project delivery.	Strengthens students' practical understanding of web development and database connectivity in software and application development.

- Images / Screenshot of the practice:

Welcome, Suryaprabha!

Department: CSE
Register Number: CSE001
Overall CGPA: 8.05 / 10

Semester-wise Marks & GPA

Semester 1 (GPA: 8.33/10)

Subject	IA1	IA2	Total	Semester Grade
Tamil	64	72	68	A
English	74	70	72	B+
Chemistry	52	84	68	A
C Programming	78	68	73	A+
Mathematics	60	67	63.5	O
Physics	74	63	68.5	A

Semester 2 (GPA: 8.6/10)

Subject	IA1	IA2	Total	Semester Grade
Mathematics I	35	37	36	O
Tamil	30	30	30	A
English	36	38	37	B
C Programming	25	30	27.5	O
Physics	28	32	30	A+

Staff Dashboard - Student Records

Search by Name, Register No, Dt

ID	Name	Register No	Department	CGPA	Action
1	Suryaprabha	CSE001	CSE	8.05	View
2	Ananya	CSE002	CSE	8.38	View
3	Rahul	CSE003	CSE	7.95	View
4	Priya	CSE004	CSE	7.9	View

[Upload Next Semester Marks \(Excel/CSV\)](#)

Figure 1



Figure 2

Project Based Learning Activity

Reflective Critique

Learning Outcomes

- Apply PHP and MySQL concepts practically to develop a functional, database-driven web application.
- Understand and implement database connectivity in PHP for CRUD operations (Create, Read, Update, Delete).
- Design and develop front-end interfaces using HTML and CSS integrated with backend functionality.

- Perform data validation and user authentication, ensuring secure and accurate data handling.
- Students gained confidence in developing and presenting mini-projects using open-source technologies, preparing them for future capstone or industry projects.

Benefits of the Practice

- The project prepared students for real-world web development tasks, improving employability skills.
- Successfully implementing a functional system increased students' confidence in coding, testing, and presenting technical solutions.
- Students could apply web technology concepts such as CRUD operations, sessions, and form handling in a practical scenario.

Feedback of the Practice

- The Project-Based Learning activity helped students understand database connectivity in PHP practically.
- It improved their teamwork, problem-solving, and coding skills through real-world implementation.
- Students appreciated the hands-on approach, which bridged the gap between theory and application.

Challenges Faced in Implementation

- Initially, students faced difficulty in establishing PHP–MySQL connectivity due to configuration issues (e.g., localhost setup and database permissions).
- Time management was challenging since the project was completed within two weeks, requiring consistent effort and coordination.
- Debugging runtime errors and ensuring data validation required additional research and trial-and-error learning.

References

<https://www.powerschool.com/blog/project-based-learning-benefits-examples-and-resources/>
<https://itali.uq.edu.au/teaching-guidance/teaching-practices/active-learning/project-based-learning>
<https://ctl.columbia.edu/resources-and-technology/resources/project-based-learning/>